

SRIRATHI MUTHURAMAN

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Srirathi Muthuraman is an aspiring Biotechnologist, looking for an opportunity to develop her research perspective to international standards and work in a challenging environment that offers a platform to upgrade her skills while leveraging the same for the benefit of the organization. She is an enthusiastic person with an immense drive to promote better standards of life through the efficient utilization of science and technology.

EDUCATION

2021– 2023

MASTER OF TECHNOLOGY, BIOTECHNOLOGY

B. S. ABDUR RAHMAN INSTITUTE OF SCIENCE AND TECHNOLOGY, CHENNAI

- CGPA – 9.29 (Till 3rd Semester)

2017 – 2021

BACHELOR OF TECHNOLOGY, BIOTECHNOLOGY

DR. MGR EDUCATIONAL AND RESEARCH INSTITUTE, CHENNAI

- CGPA – 9.51

INTERNSHIP AND TRAINING

JANUARY 2023 – ONGOING

MASTER THESIS STUDENT, INDIAN INSTITUTE OF SCIENCE, BANGALORE

“Elucidating the role of TRIM36 in Zebrafish Early Embryogenesis” under the guidance of Prof. Upendra Nongthomba, Developmental & Biomedical Genetics Laboratory, IISC

AUGUST 2022

HANDS-ON TRAINING, CSIR – CCMB, HYDERABAD

- Zebrafish Husbandry
- Microinjection
- Gene Editing Techniques
- Imaging

‘Microinjection technique to produce transgenic generations in Zebrafish’ under the guidance of Dr. Megha Kumar, CCMB

AUGUST 2022

ONLINE HANDS-ON TRAINING, CIIC & SIMBIOEN LABS & SCIENTIFIC SERVICES

ADVANCED BIOINFORMATICS (PLANT TO DRUG)

- i. Network Pharmacology Analysis
- ii. Molecular Docking
- iii. Simulation using Desmond and Mechanism Analysis

JANUARY 2022 – FEBRUARY 2022

INTERSHIP TRAINING, CAMBRIONICS LIFE SCIENCE., CHENNAI

- i. Zebrafish Handling and Developmental Toxicology
- ii. Molecular Biology
- iii. Bioinformatics and Omics Approaches

OCTOBER 2020 – JANUARY 2021

STUDENT INTERN, WAVE CODE LOGIX PRIVATE LTD., CHENNAI

R&D- Coding & Innovations Department

DECEMBER 2019

INTERN, MADURAI DISTRICT CO-OPERATIVE MILK PRODUCERS' UNION LTD., MADURAI

Quality assessment for raw and processed milk products

MAY 2019

INTERSHIP TRAINING, DOLPHIN FOODS INDIA LTD., HYDERABAD

- i. Quality Control Department and Processing Unit
- ii. Chemical and Biological Analysis
- iii. Basic Principles of Manufacturing Process

PROJECTS

PROJECT I: 'Probiotic Instant Millet Drink - An Advancement in Millet Food Preservation' under the guidance of Dr. Arun Kumar at Dr. MGR Educational and Research Institute (January 2021 – May 2021)

- Physical and Biochemical analysis of millet grains
- Culturing and Processing of the microorganism *L. fermentum*
- Fermentation Process
- Lyophilization
- Final Product Testing – Biochemical tests
- Product Enhancing Techniques

Incubated under the scheme of NewGen IEDC, granted seed-funding of 2,50,000 INR by the Department of Science and Technology, Government of India.

PROJECT II: 'Developmental toxicology for the Novel Breast cancer Drug' under the guidance of Dr. Ashok Kumar Pandurangan at B. S. Abdur Rahman Crescent Institute of Science and Technology, Chennai, India (June 2022)

- Zebrafish breeding and egg collection
- Determination of LC50 value
- Observation of abnormalities in Zebrafish embryos at different dosages of Drug

PROJECT III: 'DSS induced Intestinal Inflammation in Zebrafish Larvae and Treatment with a Novel Drug' under the guidance of Dr. Ashok Kumar Pandurangan at B. S. Abdur Rahman Crescent Institute of Science and Technology, Chennai, India (October 2022 – December 2022)

- Synthesis of Drug for Ulcerative Colitis (UC) treatment
- Drug Biochemical Analysis
- Induction of UC to Zebrafish Larvae using Dextran Sulfate Sodium
- Treatment with the novel drug at different concentrations to determine its activity against UC

TECHNICAL EXPERTISE

Microbiology

Isolation and Identification, Inoculation Techniques, Microscopy

Molecular Biology

Isolation of DNA, PCR Techniques

Biochemistry

Protein isolation, SDS PAGE, Western Blot, Chromatographic Techniques, Immunoassays

Animal Studies

Zebrafish - Maintenance, Breeding, Dechoronation, Toxicology studies, Disease Modelling, Microinjection

CERTIFIED COURSES

NATIONAL PROGRAMME ON TECHNOLOGY ENHANCED LEARNING (NPTEL)

- NPTEL - CELL CULTURE TECHNOLOGIES - 87% (FEBRUARY - APRIL 2020)
- NPTEL - BIOMEDICAL NANOTECHNOLOGY - 62% (SEPTEMBER - OCTOBER 2020)

COURSERA

- Algae Biotechnology – 98.15% (MARCH 2022)
- The Science of Stem Cells – 100% (AUGUST 2022)
- Introduction to the Biology of Cancer – 97.67 (OCTOBER 2022)
- Understanding Cancer Metastasis – 98.58 (DECEMBER 2022)

CONFERENCES AND WORKSHOPS

- ICANNOCB-22 - 3rd International Conference on “The Applications of Natural Compounds, Nanomaterials, Oncolytics in Cancer Biology and Biotechnology” organized by the School of Life Sciences - BSACIST and Association of Cancer Education and Research (ACER), in association with Purdue University, USA (27-28th October 2022). **BEST ORAL PRESENTATION AWARD**

- Facilitated 'Hands-on training on Extraction and Phytochemical Analysis of Natural Compounds – invivo and insilco Approach" – SLS, BSACIST (21-23rd September 2022)
- Webinar on "CRISPR-CAS system and its application: A Modern Era Tool in Molecular Biology" – Dr. V Naveen Kumar, Founder & MD, Immungenix Biosciences Pvt Ltd (21st November 2020).
- National Level Conference SARVASASTRA 2019 - Department of Biotechnology, Dr. MGR Educational & Research Institute (28-29th August 2019)
- FOLDSCOPE: Outreach and Applications; Workshop on handling Foldscope, a pocket microscope (24-25th July 2019).
- BIOZONE 2019 A National level Technical Symposium - Dr. MGR Educational & Research Institute (28-29th March 2019).
- HETICON' 18 - An International Conference on "The role of Technology in Shaping Future Healthcare" - Dr. MGR Educational & Research Institute, Chennai (30-31st October 2018).

REFERENCE

Dr. Ashok Kumar Pandurangan

Associate Professor

B. S. Abdur Rahman Crescent Institute of Science and Technology, Chennai

Phone: +91 82481 45481

Email: ashokkumar.sls@crescent.education

Website: <https://crescent.education/university/schools/school-of-life-sciences/faculty/dr-p-ashok-kumar/>

Mr. Sudharsan Varatharajan

CEO, Co-founder

Biversity (TABS Learning private limited)

Phone: +91 81447 99592

Email: sudharsan@biversity.io

Website: <https://biversity.io/>

DECLARATION

I hereby declare that the above information furnished by me is true to the best of my knowledge and belief.

Place : Chennai, India

Date : 26-02-2023

Signature

(SRIRATHI MUTHURAMAN)